

Octonions, G_2 , and Knot Topology in the Arithmetic of Order Framework: A Topological–Algebraic Reinterpretation of the Higgs Mechanism within the Standard Model

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Dedication

This work is dedicated to the foundational pioneers of octonionic and exceptional algebraic approaches to physics, whose groundbreaking insights have paved the way for modern formulations connecting algebra, topology, and fundamental interactions. As detailed in the Appendix, the profound contributions of researchers such as Catto, Chesley, Günaydin, Piron, and Ruegg continue to inspire and guide the development of the Arithmetic of Order framework.

Abstract

We present a unified framework connecting knot theory, exceptional algebra, and particle physics that provides a *topological–algebraic reinterpretation* of the Higgs mechanism within the Standard Model (SM). The SM gauge groups $U(1)$, $SU(2)$, and $SU(3)$ arise naturally from the three Reidemeister moves on knots and braids, with a rigorous Lie-algebraic identification via Schiller’s strand–tangle model. These gauge symmetries embed into the exceptional Lie group G_2 , the automorphism group of the octonions, and are realized geometrically through coset spaces such as $G_2/(SU(2)_L \times SU(2)_R)$.

Within the Arithmetic of Order (AoO) framework, the Higgs field is reinterpreted not as a fundamental scalar particle but as an *internal algebraic phase transition* on this coset manifold. This phase transition triggers symmetry breaking, producing W and Z boson excitations consistent with collider data. The framework yields a fully computable, discrete physical model that recasts the SM as an inevitable consequence of topology, exceptional algebra, and discrete geometry.

1 Introduction

The Standard Model (SM) successfully describes electromagnetic, weak, and strong interactions via the gauge groups $U(1)$, $SU(2)$, and $SU(3)$. Yet the *origin* of these symmetries and the nature of the Higgs mechanism remain foundational open questions.

This work develops a conceptually novel, mathematically rigorous framework that:

- Roots SM gauge symmetries in knot theory, identifying the three Reidemeister moves as generators of $U(1)$, $SU(2)$, and $SU(3)$,
- Embeds these symmetries within the exceptional Lie group G_2 , the automorphism group of the octonions,
- Reinterprets the Higgs field as an *internal algebraic phase transition* on the coset space $G_2/(SU(2)_L \times SU(2)_R)$,
- Grounds all constructions in the Arithmetic of Order (AoO) framework, which models the universe as a finite, computable structure.

This topological–algebraic viewpoint offers a striking alternative to conventional Higgs physics, eliminating the need for a fundamental scalar and naturally aligning with discrete computational physics.

2 Reidemeister Moves and Gauge Symmetries

Each Reidemeister move corresponds precisely to an SM gauge group:

Move	Topological Action	Gauge Group
R1 (Twist)	Single-strand rotation	$U(1)$
R2 (Poke)	Two-strand crossing	$SU(2)$
R3 (Slide)	Three-strand sliding	$SU(3)$

Schiller’s (2025) strand–tangle model rigorously constructs explicit algebraic generators for these moves, demonstrating their closure under the corresponding Lie algebras. This places the topology–gauge correspondence on firm algebraic footing.

3 Octonions and G_2 Embedding

The octonions $\mathbb{O}$ form an eight-dimensional normed division algebra whose automorphism group is the exceptional Lie group G_2 . Crucially,

- $SU(3) \subset G_2$ acts as the color symmetry,
- $SO(4) \cong (SU(2)_L \times SU(2)_R)/\mathbb{Z}_2$ encodes the electroweak symmetry.

Schiller’s algebraic generators embed explicitly into these subalgebras, unifying knot topology and SM gauge symmetries within G_2 .

4 Geometric Realization and the Higgs Field

4.1 Coset Space Structure

Within AoO, the Higgs field is interpreted as a coordinate $\phi(x)$ on the coset manifold

$$G_2/(SU(2)_L \times SU(2)_R) \cong S^6,$$

representing an *internal algebraic phase transition* or *coset twist* in the octonionic structure.

The standard Higgs doublet emerges naturally from the related coset

$$SU(3)/(SU(2)_L \times U(1)_Y) \cong \mathbb{CP}^2,$$

capturing electroweak symmetry breaking phenomenology.

4.2 Effective Lagrangian and Couplings

In terms of strand variables, the effective Higgs-sector Lagrangian is

$$\mathcal{L}_\phi = \frac{1}{2}(\partial_\mu \phi)^2 - V(\phi) + \bar{\psi}\Gamma(\phi)\psi + \dots,$$

where $V(\phi)$ encodes topological rearrangement energies and $\Gamma(\phi)$ governs couplings to the W and Z bosons.

4.3 AoO Interpretation

- The Higgs field is not fundamental but arises as an emergent algebraic phase transition within octonionic coset geometry.
- The observed 125 GeV Higgs “signal” corresponds to gauge boson excitations emerging from this internal transition.
- This approach fits naturally within a fully discrete, computable physical universe described by the AoO framework.

5 Computability, Predictions, and Tests

- The AoO framework models the physical world as a finite computation, consistent with collider data including the LHC.
- No free fundamental scalar propagator is expected; only gauge boson decay products are observed.
- The framework predicts precise Higgs–gauge coupling constants derivable from topological strand data, enabling falsifiable experimental tests such as Higgs self-coupling measurements.
- It points toward potential extensions beyond the Standard Model through higher exceptional groups and discrete algebraic geometries.

6 Conclusions

This work:

1. Roots Standard Model gauge groups in fundamental knot-theoretic moves,
2. Embeds these symmetries in the exceptional Lie group G_2 via octonions,
3. Provides a novel topological–algebraic reinterpretation of the Higgs mechanism,
4. Offers a finite, computable model aligned with experimental evidence and consistent with discrete physics principles.

Future directions include exploring E_8 extensions, integrating discrete gravity into AoO, and computational implementations of quantum field theory.

Appendix: Foundational Octonionic Work

The AoO framework builds on pivotal developments in exceptional algebra and physics, notably:

- S. Catto and D. Chesley, *Twisted Octonions and Their Symmetry Groups*, *Nuclear Physics B* **331** (1989), foundational for twisted-octonion algebra under G_2 and Standard Model subgroup embeddings.
- M. Günaydin, C. Piron, and H. Ruegg, *Moufang Plane, Jordan Algebras and Octonionic Quantum Mechanics*, *Journal of Mathematical Physics* **25** (1984), pioneering connections between octonionic geometry, Jordan algebras, and quantum mechanics consistent with AoO’s discrete algebraic geometry.

References

- [1] Faysal El Khettabi, *Arithmetic of Order: A Fundamental Reformulation*, 2025. Full development of hypercomplex numbers under powersets and computable structures explained in the AoO RoadMap: <https://efaysal.github.io/HCNFEK2024FE/HypComNumSetTheGCFEKFEB2024.pdf> and the subsequent realizations in *Turing Emergence Theorem and AoO-Computable Physics*, <https://efaysal.github.io/HCNFEK2024FE/TuringA00FEKAUGUST2025.pdf>.
- [2] C. Schiller, *Testing a Model for Emergent Spinor Wave Functions Explaining Elementary Particles, Gauge Interactions and Fundamental Constants*, Motion Mountain Research, 2025.
- [3] S. Catto and D. Chesley, “Twisted Octonions and Their Symmetry Groups,” *Nuclear Physics B*, 1989.
- [4] M. Günaydin, C. Piron, and H. Ruegg, *Moufang Plane, Jordan Algebras and Octonionic Quantum Mechanics*, *Journal of Mathematical Physics* **25**, no. 8 (1984), 1447–1461.